



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

IS1103-1 – INTRODUCTION TO SOFTWARE ENGINEERING
(IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **A step in waterfall model that involves a meeting with the customer to understand the requirements.**
 - A) Implementation
 - B) Requirement Gathering
 - C) Customer review
 - D) SRS
2. **Select the odd One**
 - A) The app should be able to handle a high volume of concurrent users
 - B) The app's UI should be intuitive and easy to navigate for users with varying technical skills
 - C) The system notifies the user when a new order arrives
 - D) The system enables users to place lunch orders.
3. **A planned program which requires a definitive amount of time, effort and planning to complete.**
 - A) Project
 - B) Problem
 - C) Program
 - D) Process
4. **What is the purpose of a feasibility study in requirements engineering**
 - A) To determine the project's budget
 - B) To assess the project's viability and practicality
 - C) To evaluate the technical skills of the development team
 - D) To define the project timeline
5. **Which of the following is an example of a software maintenance activity?**
 - A) Writing new code for a feature
 - B) Conducting user training
 - C) Creating initial documentation
 - D) Fixing a bug in the existing code
6. **What is the main advantage of the Incremental Model?**
 - A) Sequential development
 - B) Minimal customer involvement
 - C) Early delivery of partial product
 - D) Rigorous planning
7. **Which of the Software process model gives explicit recognition of risk**
 - A) Waterfall Model
 - B) Incremental Development Model
 - C) Prototype Model
 - D) Spiral Development Model
8. **"Consider a system where, a heat sensor detects an intrusion and alerts the security company." What kind of a requirement the system is providing ?**
 - A) Known Requirement
 - B) Functional Requirement
 - C) Non Functional Requirement
 - D) System Requirement
9. **Select the odd One**
 - A) The system enables users to place lunch orders.
 - B) The system notifies the user when a new order arrives.
 - C) The system always responds to user clicks in less than one tenth of a second.
 - D) The system displays a list of hotel vacancies.
10. **Which of the following is the primary goal of the Requirements Engineering phase in software development?**
 - A) Writing efficient code
 - B) Defining the functionalities and features of the software
 - C) Deploying the software to users
 - D) Testing the software for bugs

11. **Agile and Waterfall are two contrasting software development life cycle (SDLC) models. What is a key difference between them?**
- A) Agile prioritizes iterative development and user feedback, while Waterfall follows a linear, sequential approach.
 - B) Agile is better suited for large, complex projects, whereas Waterfall is ideal for smaller projects.
 - C) Agile requires extensive documentation upfront, while Waterfall encourages minimal documentation.
 - D) Agile is best for projects with fixed requirements, while Waterfall is flexible for changing needs.
12. **What elements typically represent lifelines in a sequence diagram?**
- A) Data flows and control flows between objects.
 - B) User interface elements and their interactions with the system.
 - C) Objects participating in the interaction and their lifespans.
 - D) Packages and subsystems within the overall system architecture.
13. **What is the main advantage of using class diagrams in UML?**
- A) To define the sequence of steps in a specific user interaction.
 - B) To visualize the relationships and collaborations between classes.
 - C) To document the detailed implementation logic of a method.
 - D) To illustrate the different user interfaces available in a system.
14. **What kind of relationships can be represented between classes in a class diagram?**
- A) Only inheritance relationships between parent and child classes.
 - B) "Has-a" and "Uses-a" relationships only.
 - C) Inheritance, association, aggregation, and composition relationships.
 - D) Sequential interactions between classes in a specific scenario.
15. **Which of the following is NOT a core responsibility of a project manager?**
- A) Defining project scope and deliverables
 - B) Motivating and leading the project team
 - C) Managing project budget and resources
 - D) Identifying and mitigating project risks
16. **Effective teamwork in project management emphasizes:**
- A) Micromanaging individual tasks and contributions
 - B) Assigning blame for mistakes and missed deadlines
 - C) Open communication, collaboration, and shared goals
 - D) Prioritizing individual achievements over team success
17. **When prioritizing tasks during project planning, which factor is MOST important to consider?**
- A) The estimated duration of each task in isolation
 - B) The dependencies between tasks and their impact on the overall schedule
 - C) The individual preferences and skillsets of team members
 - D) The perceived difficulty or complexity of each task
18. **What is the key benefit of creating a well-defined project plan?**
- A) To impress stakeholders with detailed documentation
 - B) To guarantee a successful project outcome regardless of unforeseen circumstances
 - C) To eliminate the need for communication and collaboration within the team
 - D) To provide a clear roadmap for project execution and milestones
19. **During project execution, a critical team member unexpectedly falls ill and requires extended leave. What is the MOST important action for the project manager to take in this situation?**
- A) Reprimand the team member for missing work
 - B) Assign all the absent team member's tasks to another team member without adjustment
 - C) Communicate openly with the team and stakeholders, assess the impact on the project, and develop a mitigation plan.
 - D) Ignore the situation and hope the team member recovers quickly
20. **Agile project management methodologies emphasize:**
- A) Rigid adherence to pre-defined plans and timelines.
 - B) Iterative development, continuous feedback, and adaptation
 - C) Limited team involvement and a focus on top-down decision-making.
 - D) Extensive documentation and detailed upfront planning.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Company XYZ wants to deploy a new E-Commerce Platform, explain the requirement engineering phase activities with respect to it.	8	L*2	1	2
b)	Explain the concept of software processes and software process models.	8	L2	2	1
2. a)	A software development team is working on a safety-critical application for autonomous vehicles. They are following the V-Model (Verification and Validation Model) for development. During the testing phase, they discover a critical defect that could impact vehicle safety. How should they address this issue within the V-Model framework?	8	L3	1	2
b)	Explain the activities involved in software requirement engineering, software development, testing and evolution.	8	L2	2	1
3. a)	Describe the waterfall, incremental and iterative models and architectural design in implementing the software.	8	L2	2	1
b)	Illustrate Functional and Non-functional requirement related to a banking application.	8	L2	2	2
Unit – II					
4. a)	Create a simple data flow diagram for an online food ordering system, highlighting the data sources, data flow, and data transformations involved.	8	L3	2	2
b)	How can Agile development be effective in managing the ever-changing requirements in software projects?	8	L1	3	3
5. a)	You are developing a library management system. Draw a use case diagram depicting the interaction between the librarian and the system.	8	L2	4	3
b)	How can Agile practices be adapted to suit different project types and sizes?	8	L1	4	3
6. a)	You are designing an e-commerce platform. How can state diagrams help you visualize the different states of an order in the system and the transitions between them?	8	L2	3	3
b)	Describe the process of planning and estimating work in an Agile sprint.	8	L1	4	3
Unit – III					
7. a)	Discuss in detail the significance of risk evaluation.	8	L2	5	2
b)	Explain with an example how critical path can be identified in precedence networks?	8	L2	5	3
8. a)	Explain any two network planning models used in Software Project Management.	8	L2	5	3
b)	Explain the need to create cohesive project groups and its advantages.	8	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
